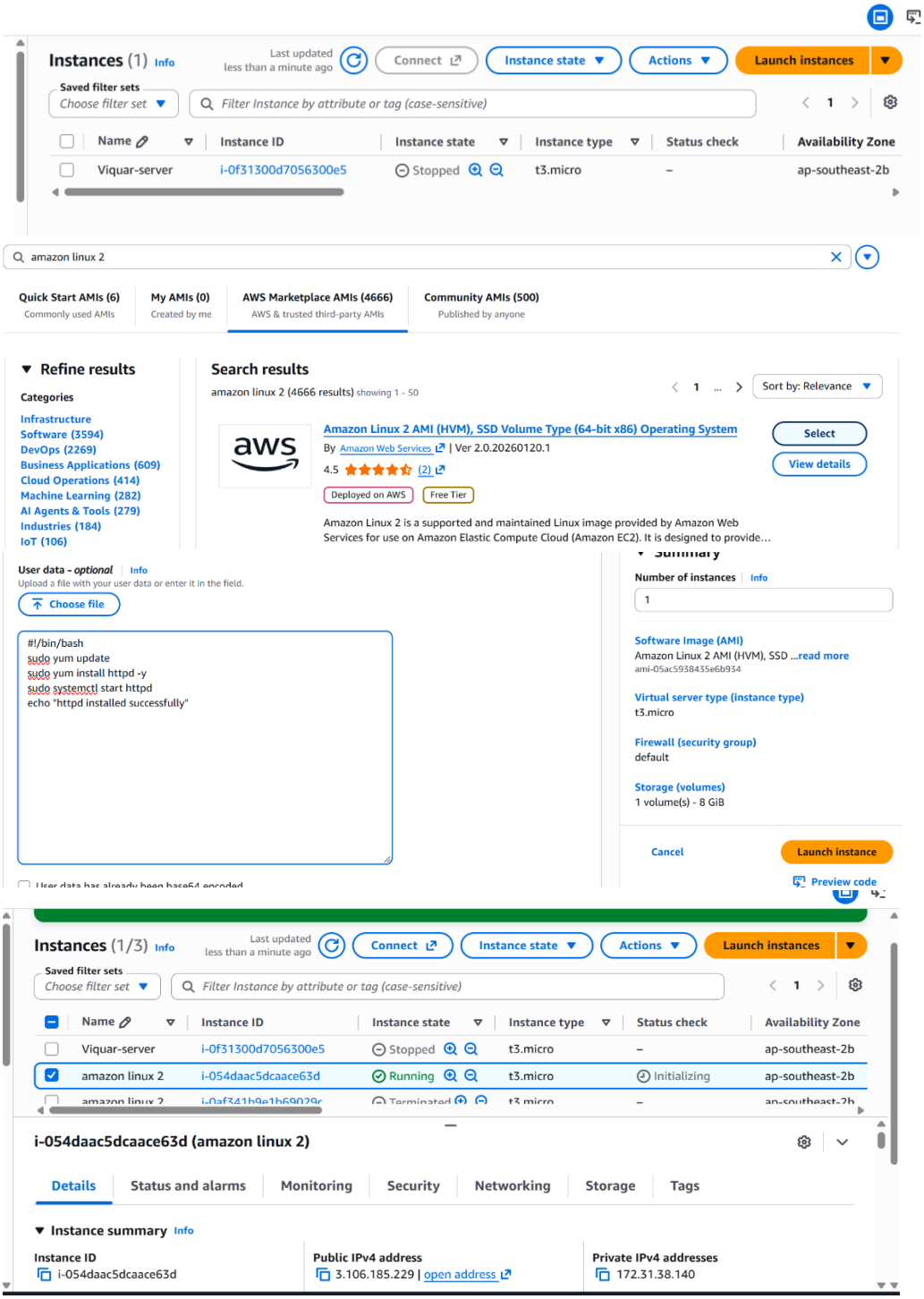
1. **Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.**

Steps : Searched for amazon Linux 2 and get in AWS marketplace AMIs
selected and continue with key pair then security group default
Selected advance details and in that below see option user data where I have written a bash script
#!/bin/bash
Sudo yum update
Sudo yum install httpd -y
Sudo systemctl start httpd
Then launch the instance → get to gitbash and enter ssh -I key.pem ec2-user@public-ip

results below:

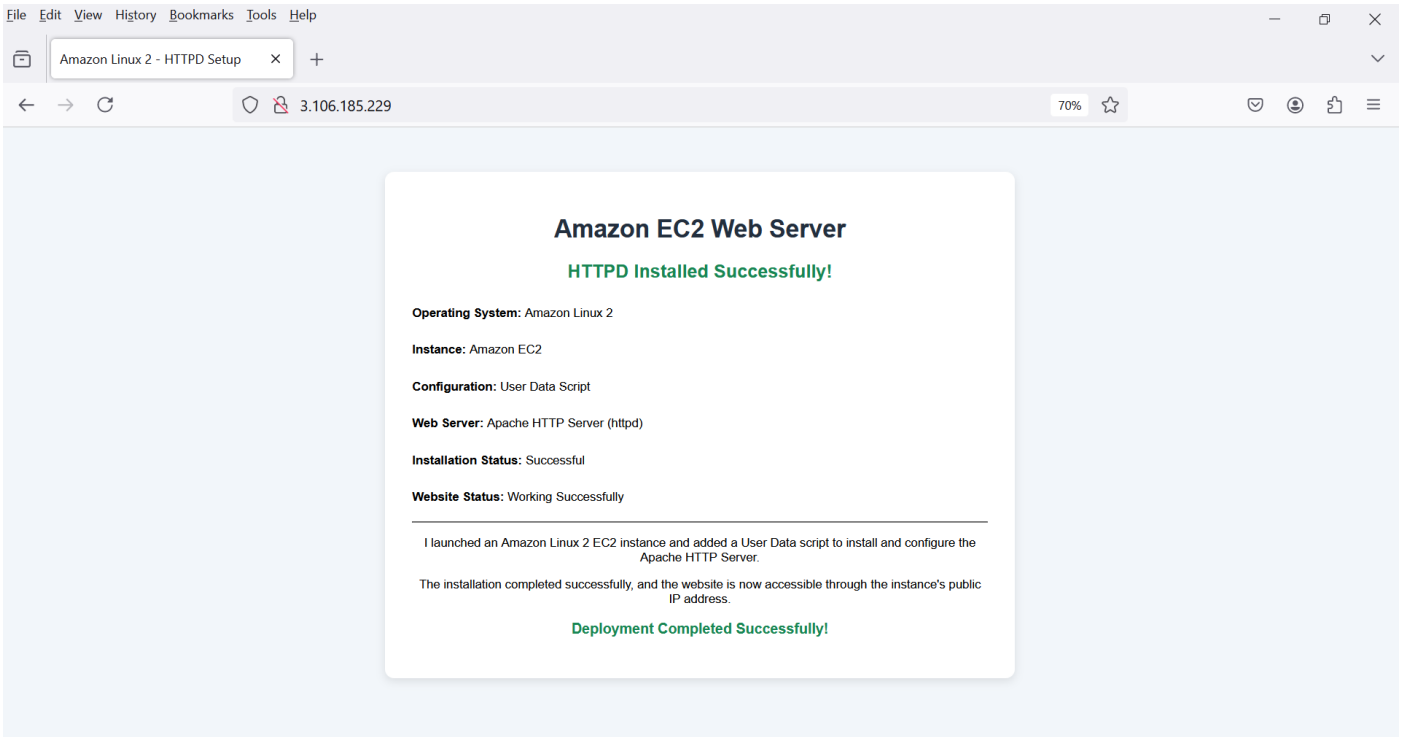


ec2-user@ip-172-31-38-140:~

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ ssh -i key.pem ec2-user@3.106.185.229
The authenticity of host '3.106.185.229 (3.106.185.229)' can't be established.
ED25519 key fingerprint is: SHA256:NcsmK28+Gv7cMpWZ3XjGzRV2ZhBemBk7UN72T+OeC4c
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.106.185.229' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2
  ~~  \_#####\
  ~~     \###|       AL2 End of Life is 2026-06-30.
  ~~       \#/ ___
   ~~       V~' '->
    ~~~         /    A newer version of Amazon Linux is available!
      ~~._.   _/
         _/ _/       Amazon Linux 2023, GA and supported until 2028-03-15.
       _/m/'           https://aws.amazon.com/linux/amazon-linux-2023/

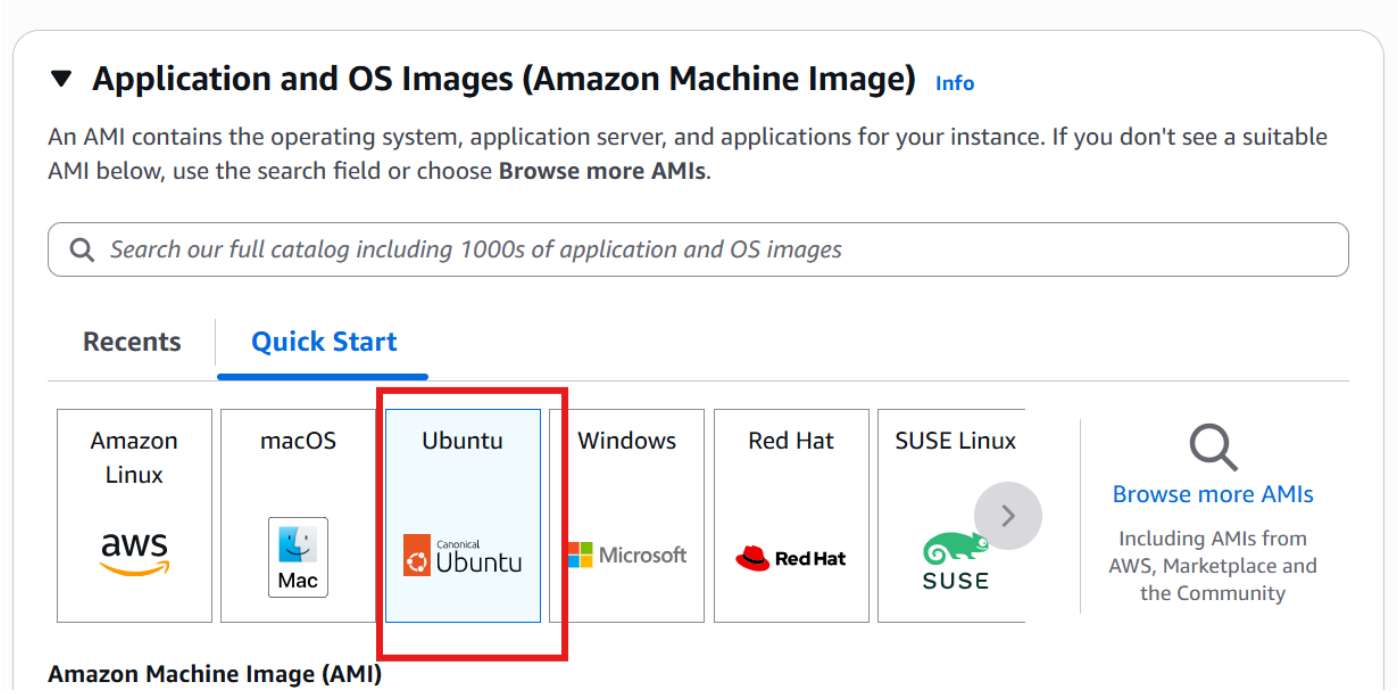
16 package(s) needed for security, out of 25 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-38-140 ~]$ systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
   Active: active (running) since Wed 2026-10-28 12:26:08 UTC; 54s ago
     Docs: man:httpd.service(8)
 Main PID: 5883 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
   CGroup: /system.slice/httpd.service
           ├─5883 /usr/sbin/httpd -DFOREGROUND
           ├─5885 /usr/sbin/httpd -DFOREGROUND
           ├─5886 /usr/sbin/httpd -DFOREGROUND
           ├─5887 /usr/sbin/httpd -DFOREGROUND
           ├─5888 /usr/sbin/httpd -DFOREGROUND
           └─5889 /usr/sbin/httpd -DFOREGROUND

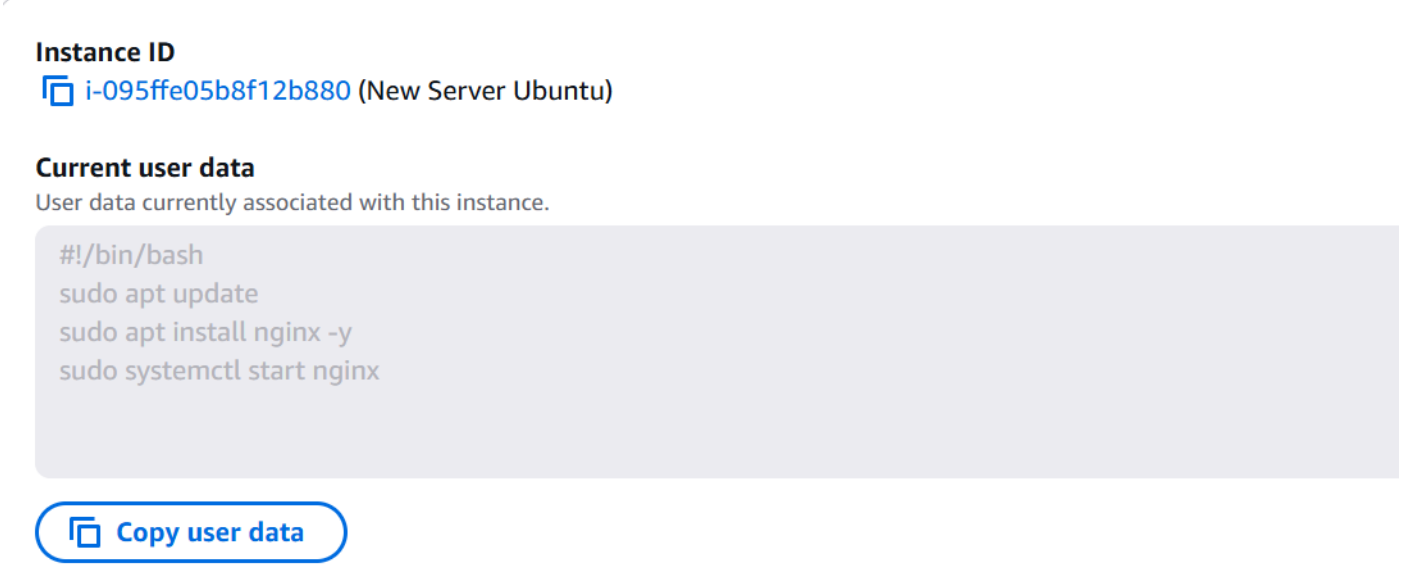
Oct 28 12:26:07 ip-172-31-38-140.ap-southeast-2.compute.internal systemd[1]: Starting The Apache HTTP Server...
Oct 28 12:26:08 ip-172-31-38-140.ap-southeast-2.compute.internal systemd[1]: Started The Apache HTTP Server.
[ec2-user@ip-172-31-38-140 ~]$
```

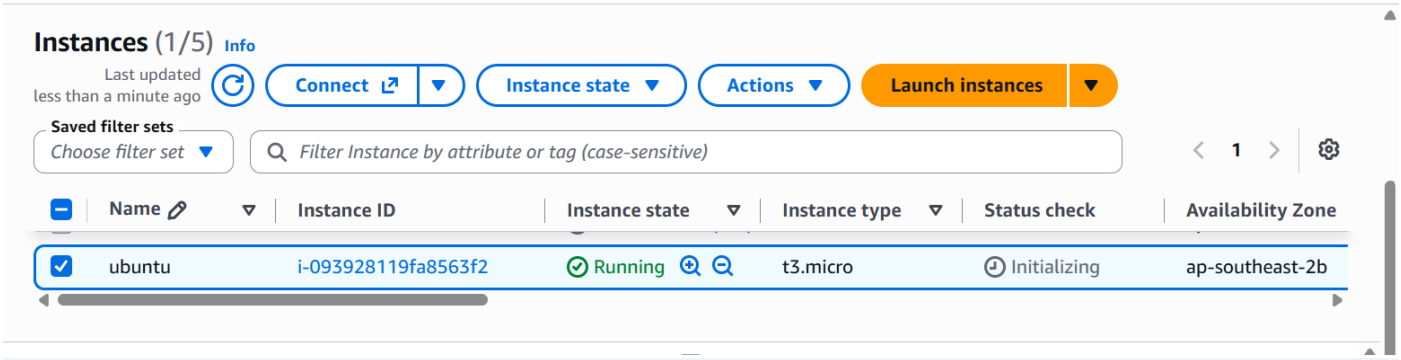


2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

Steps : selected ubuntu image
selected and continue with key pair then security group default
Selected advance details and in that below see option user data where I have written a bash script
#!/bin/bash
Sudo apt update
Sudo apt install nginx -y
Sudo systemctl start nginx
Then launch the instance → get to gitbash and enter ssh -I key.pem ubuntu@public-ip







ubuntu@ip-172-31-40-220: ~

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ ssh -i "key.pem" ubuntu@ec2-3-107-153-232.ap-southeast-2.compute.amazonaws.com
The authenticity of host 'ec2-3-107-153-232.ap-southeast-2.compute.amazonaws.com (3.107.153.232)' can't be established.
ED25519 key fingerprint is: SHA256:O7DRy6XOoLzW3cHDFVOJX6jOCAL6+ga7OO3FUou8N4M
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:17: 13.210.3.241
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-107-153-232.ap-southeast-2.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Oct 28 13:12:22 UTC 2026

  System load:  0.88              Temperature:           -273.1 C
  Usage of /:   30.7% of 6.61GB   Processes:             124
  Memory usage: 26%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.40.220


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
```

```
ubuntu@ip-172-31-35-254:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-28 13:33:26 UTC; 26s ago
 Invocation: af5dcef81f4a4223a932d7831bcc894c
       Docs: man:nginx(8)
    Process: 1466 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1471 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1547 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 6.6M)
        CPU: 55ms
     CGroup: /system.slice/nginx.service
             ├─1547 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1549 "nginx: worker process"
             └─1553 "nginx: worker process"
```

3. Launch one Windows server and install Tomcat on Windows.

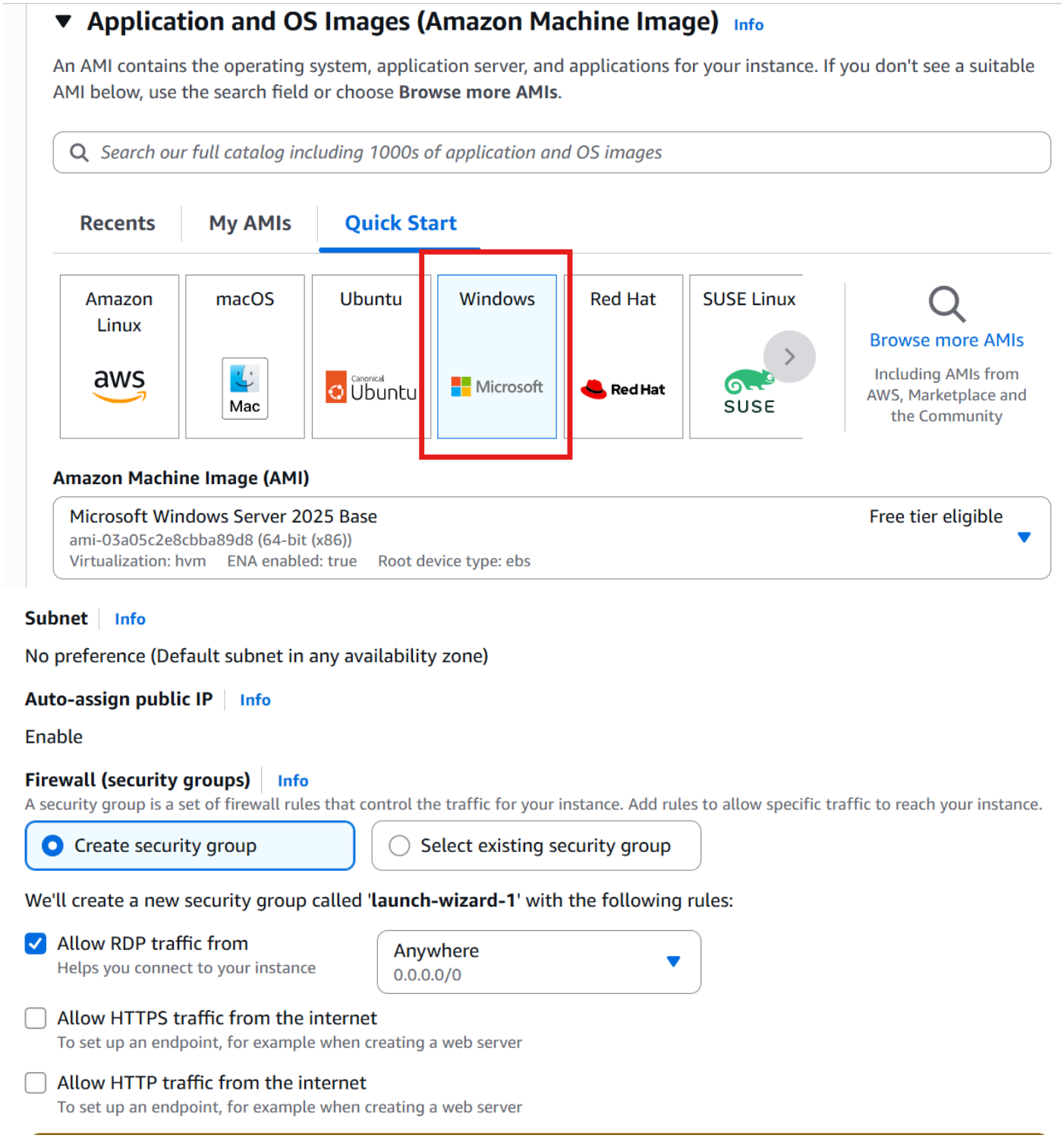
Steps: Open AWS and select windows

Security group select Allow RDP traffic → launch instance → go to connect and decrypt password

Open Remote desktop connection → paste the link copied from instance → Type user and password

Click on connect it will open GUI of Microsoft windows

Open CMD and install java and tomcat → go to browser and type localhost:8080



How to connect

1. Open or install a remote desktop client
2. Download the RDP shortcut file
3. Open the RDP client and run the shortcut file
4. Retrieve password
5. Enter username and password
6. Connect

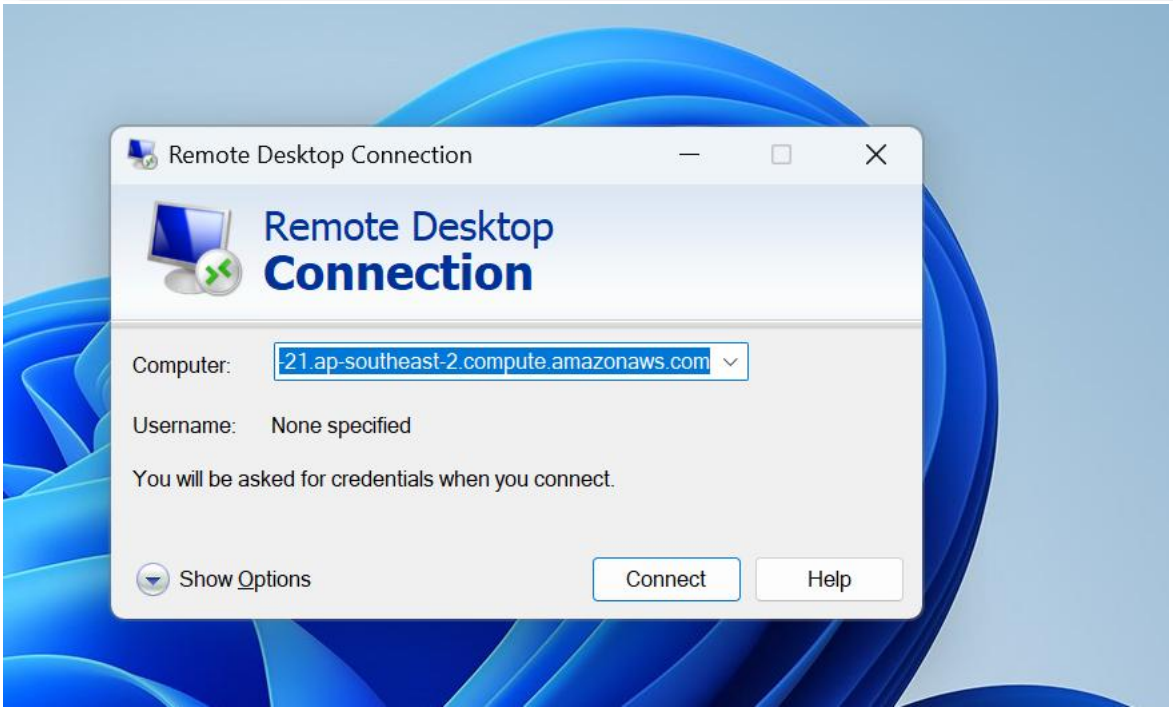
RDP shortcut file
Download

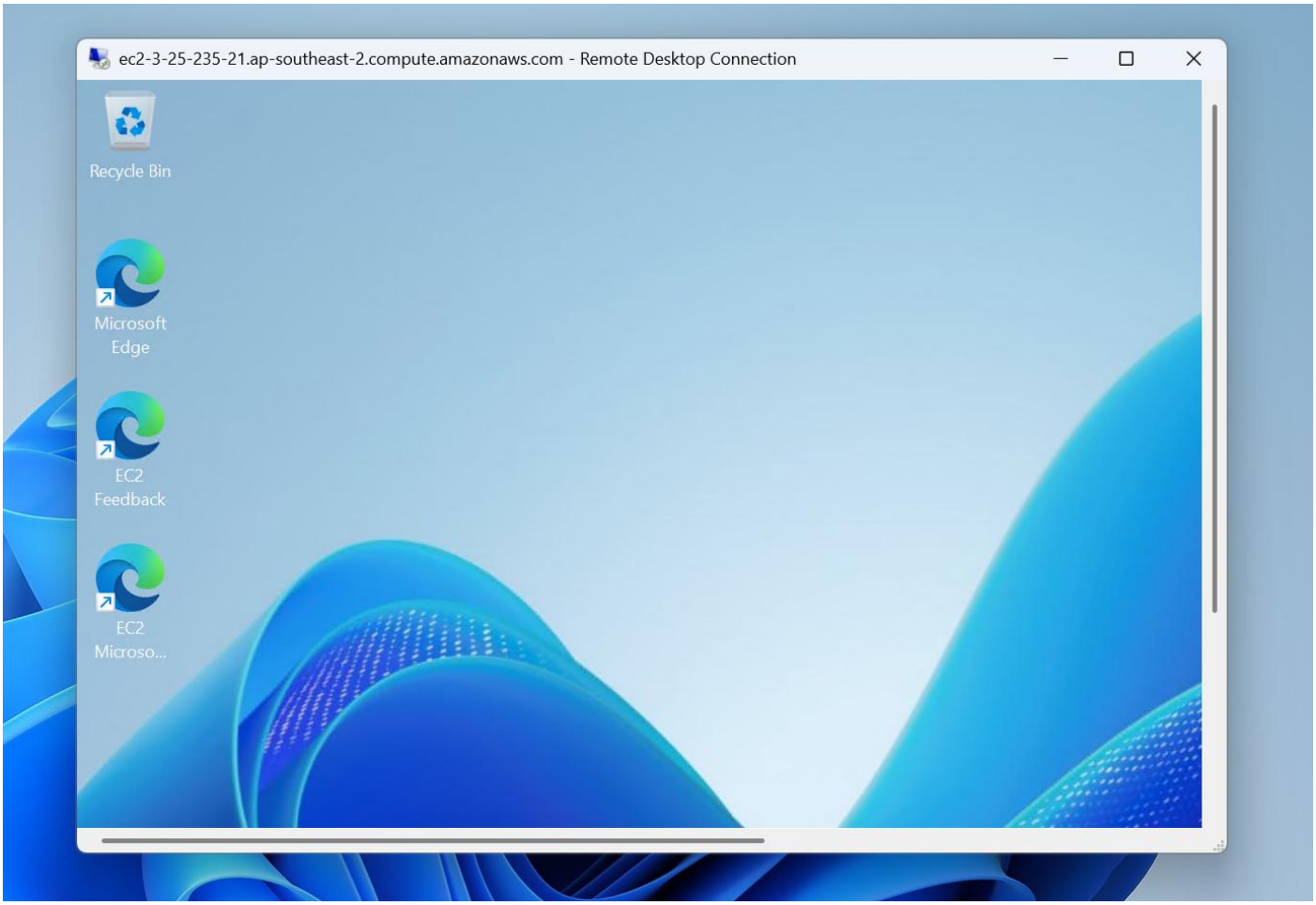
Public DNS
ec2-3-25-235-21.ap-southeast-2.compute.amazonaws.com

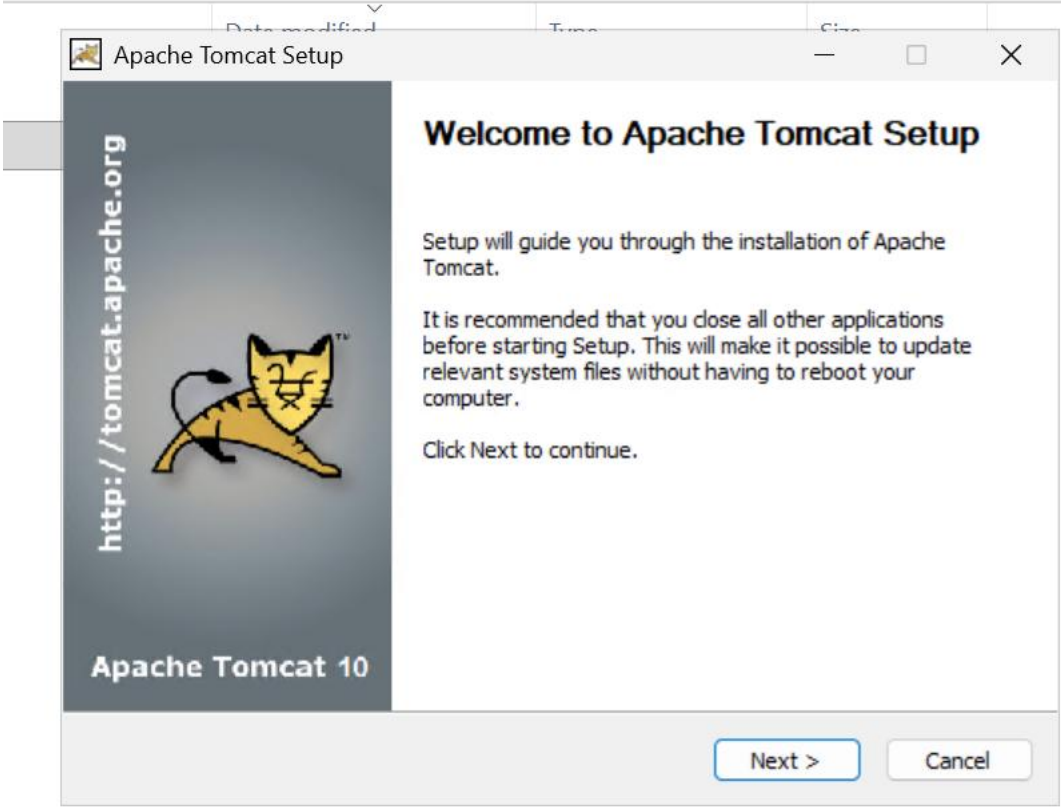
Username
Administrator

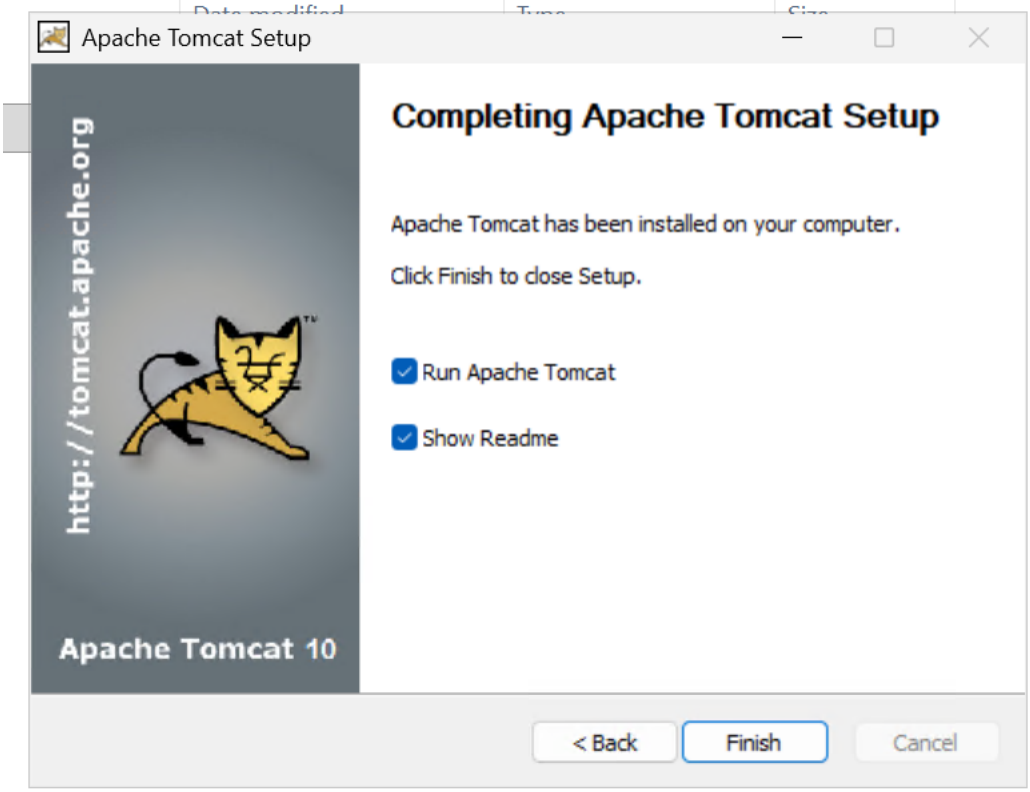
Password
Get Windows password

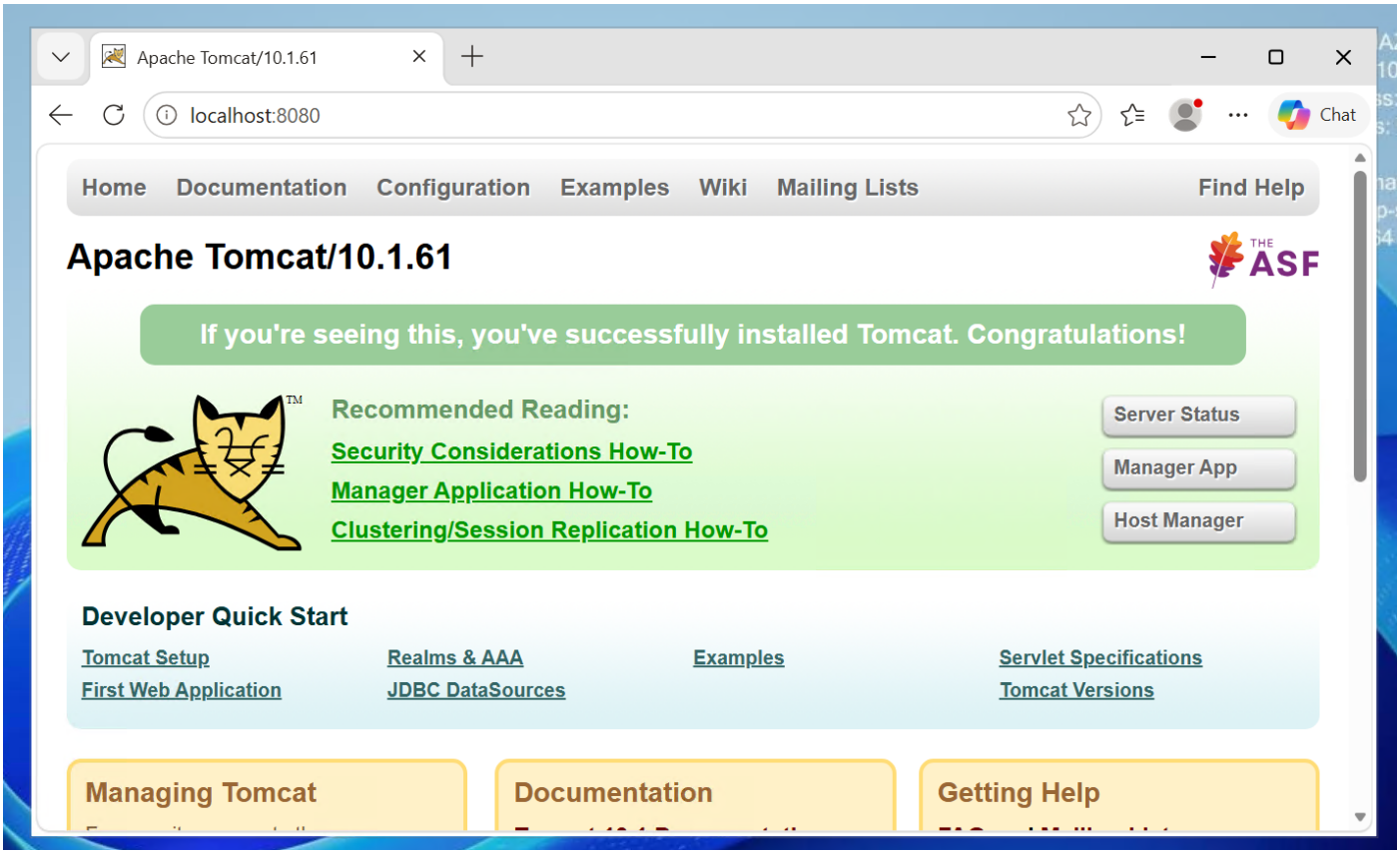
If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.




ec2-3-25-235-21.ap-southeast-2.compute.amazonaws.com - Remote Desktop Connection
Recycle Bin
Microsoft Edge
EC2 Feedback
EC2 Microso...


Apache Tomcat Setup
http://tomcat.apache.org
Apache Tomcat 10
Welcome to Apache Tomcat Setup
Setup will guide you through the installation of Apache Tomcat.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel

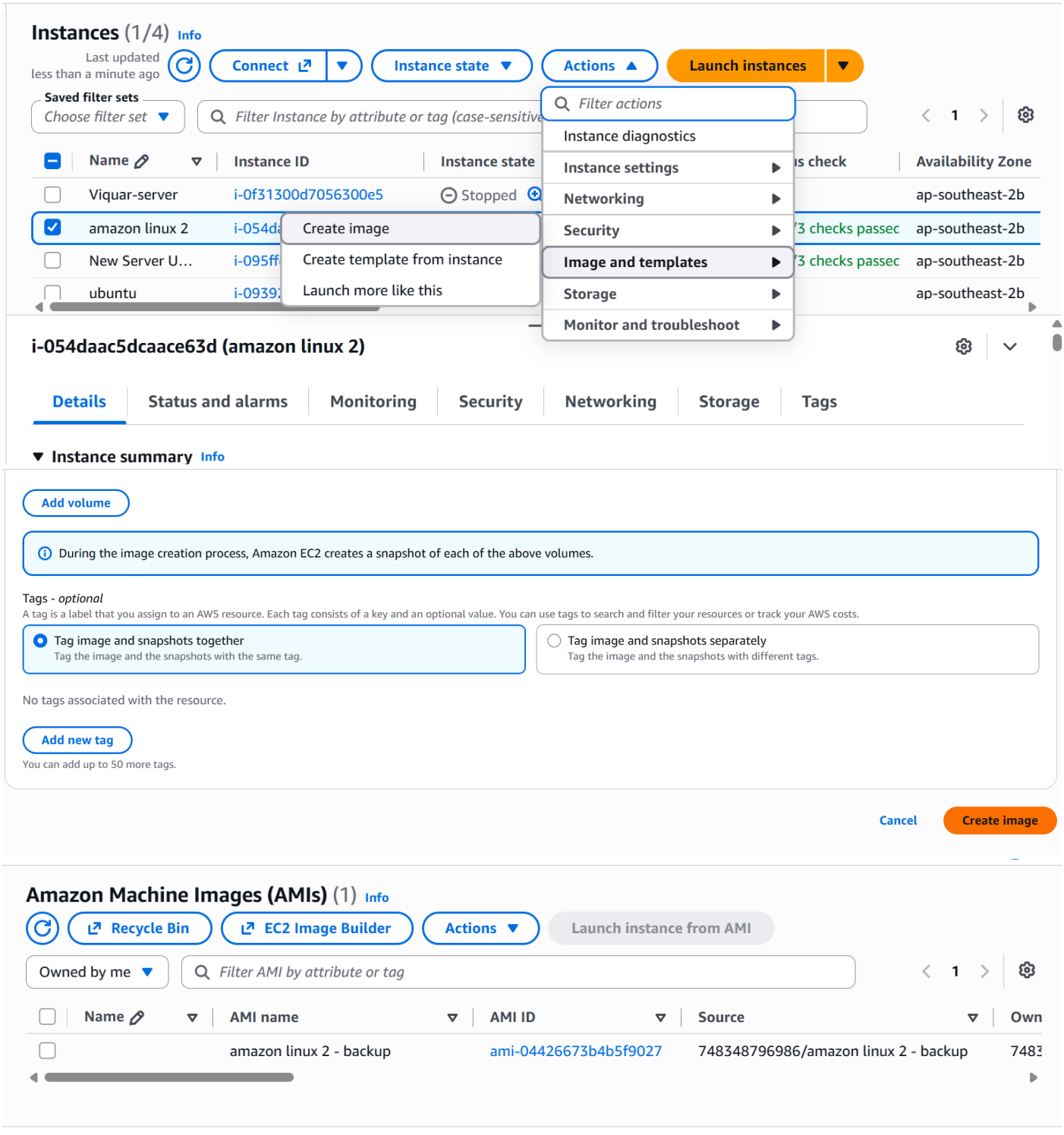

Apache Tomcat Setup
http://tomcat.apache.org
Apache Tomcat 10
Completing Apache Tomcat Setup
Apache Tomcat has been installed on your computer.
Click Finish to close Setup.
Run Apache Tomcat
Show Readme
< Back
Finish
Cancel


Apache Tomcat/10.1.61
localhost:8080
Chat
Home
Documentation
Configuration
Examples
Wiki
Mailing Lists
Find Help
Apache Tomcat/10.1.61
THE ASF
If you're seeing this, you've successfully installed Tomcat. Congratulations!
Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To
Server Status
Manager App
Host Manager
Developer Quick Start
Tomcat Setup
First Web Application
Realms & AAA
JDBC DataSources
Examples
Servlet Specifications
Tomcat Versions
Managing Tomcat
Documentation
Getting Help

4. Take a snapshot of the instance created in Task 1.

Open instance dashboard → click on ▼ down arrow beside launch instance

List will open → select option **image and templates → create image** tag image and snapshot same time



5. Assign passwordless authentication for the EC2 created in Task 2.

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ ssh ubuntu@54.252.229.47
The authenticity of host '54.252.229.47 (54.252.229.47)' can't be established.
ED25519 key fingerprint is: SHA256:nO4iaV1OfV3dKuVKwdnOEj6enDUkADfDS2dG5tgIdqU
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:22: ec2-54-252-229-47.ap-southeast-2.compute.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.252.229.47' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Oct 28 14:03:18 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   33.9% of 6.61GB   Processes:             121
  Memory usage: 26%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.35.254


Expanded Security Maintenance for Applications is not enabled.

37 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed Oct 28 14:02:15 2026 from 103.203.172.230
```

UCRT64:/c/Users/User/Downloads

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ ssh-key
bash: ssh-key: command not found

User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/User/.ssh/id_ed25519):
/c/Users/User/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/c/Users/User/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/User/.ssh/id_ed25519
Your public key has been saved in /c/Users/User/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:8UKUz379AApryjEUdCltPAA+eaOPfxB2iovEuCW189A User@DESKTOP-2QHC9VQ
The key's randomart image is:
+--[ED25519 256]--+
|     .oo+oo      |
|    . .ooB       |
|     + o+o+      |
|  .  ++oooo .    |
| + o.+ +S+.. o   |
|o B Eo= o.o . o  |
| = =.o.*    .  o |
|. . o.o .       .|
|      ..         |
+----[SHA256]-----+

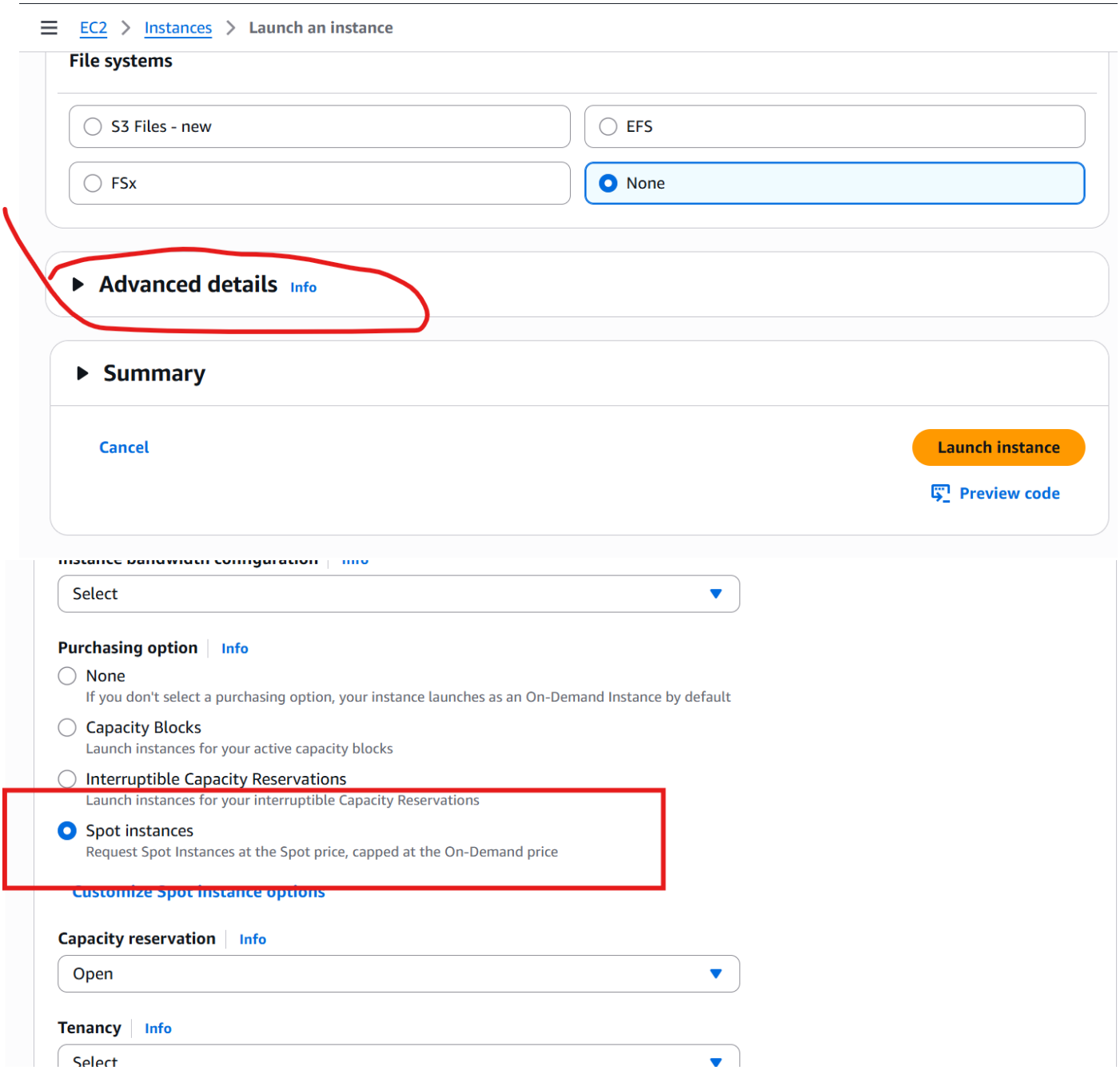
User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$ cat  /c/Users/User/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICByPYRh89oLdB7WrzSpbFNi4YzXobNIi47qA65Z22H3 User@DESKTOP-2QHC9VQ

User@DESKTOP-2QHC9VQ UCRT64 ~/Downloads
$
```

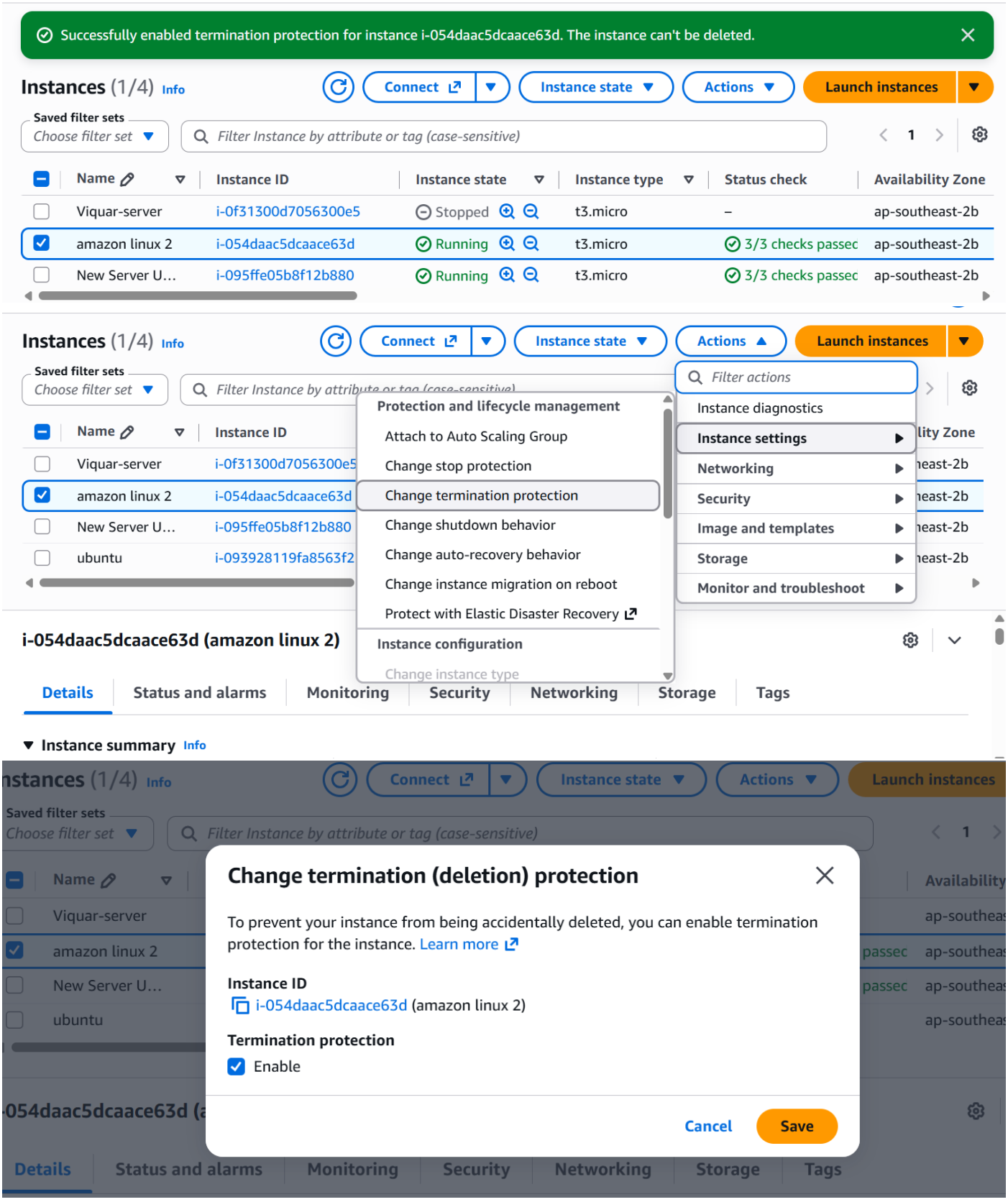
ubuntu@ip-172-31-35-254: ~/.ssh

```
ubuntu@ip-172-31-35-254:~$ cd .ssh
ubuntu@ip-172-31-35-254:~/.ssh$ ls
authorized_keys
ubuntu@ip-172-31-35-254:~/.ssh$ sudo vi authorized_keys
ubuntu@ip-172-31-35-254:~/.ssh$ sudo vi authorized_keys
ubuntu@ip-172-31-35-254:~/.ssh$ sudo nano authorized_keys
ubuntu@ip-172-31-35-254:~/.ssh$ sudo systemctl restart ssh
ubuntu@ip-172-31-35-254:~/.ssh$
```

6. Launch any EC2 using the spot purchasing option.



7. Enable termination policy on the EC2 created in Task 2.



8. Launch one EC2 using AWS CLI.

Steps : go to CLI and check for aws version → aws –version

Then write below comment

aws ec2 run-instances \

> --image-id ami-06aad1f84b1a6bce9 \

> --instance-type t3.micro \

> --key-name "key" \

> --region ap-southeast-2

Failed to launch instance because of my aws account.

